www.tntextbooks.in

short fibres coming out of the cell body are called **dendrites,** which transmit impulses towards the cell body. The cell body and the dendrites contain cytoplasm and granulated endoplasmic reticulum called **Nissl's granules.**

An axon is a long fibre that arises from a cone shaped area of the cell body called the **Axon hillock** and ends at the branched distal end. Axon hillock is the place where the nerve impulse is generated in the motor neurons. The axon of one-neuron branches and forms connections with many other neurons. An axon contains the same organelles found in the dendrites and cell body but lacks Nissl's granules and Golgi apparatus.

The axon, particularly of peripheral nerves is surrounded by **Schwann cells** (a type of glial cell) to form myelin sheath, which act as an insulator. **Myelin sheath** is associated only with the axon; dendrites are always non-myelinated. Schwann cells are not continuous along the axon; so there are gaps in the myelin sheath between adjacent Schwann

The longest cells in the human body are the **neurons**. The **longest** axons in the **human body**, for example, are those of the sciatic nerve, which run from the base of the spine to the big toe of each foot. These single-cell fibers may extend a meter or even longer. The axons of the inter neurons in the CNS are the shortest.

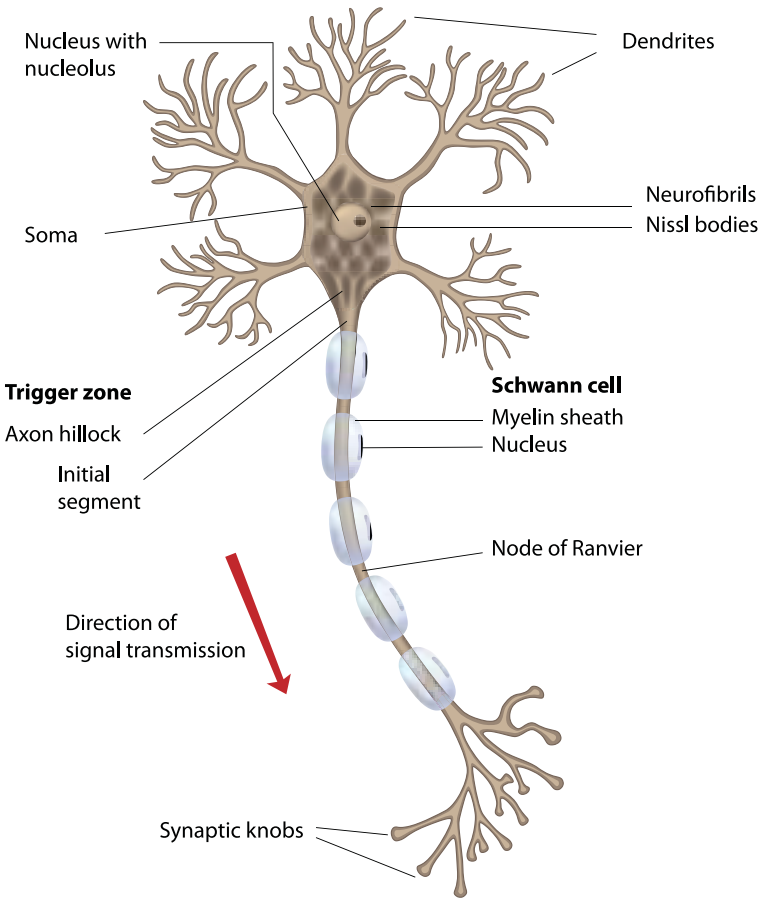


Figure 10.1 Neuron